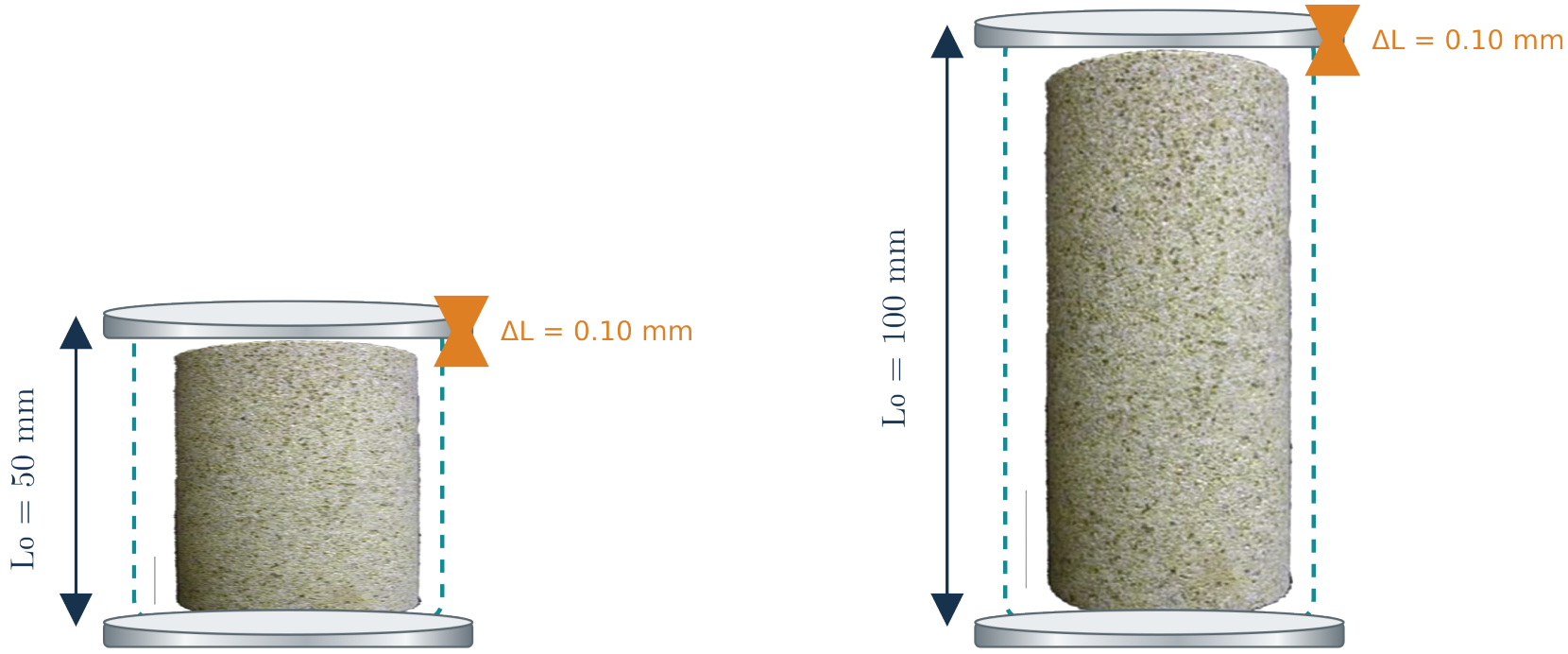
# Does equal shortening mean equal strain ?

Activity 2

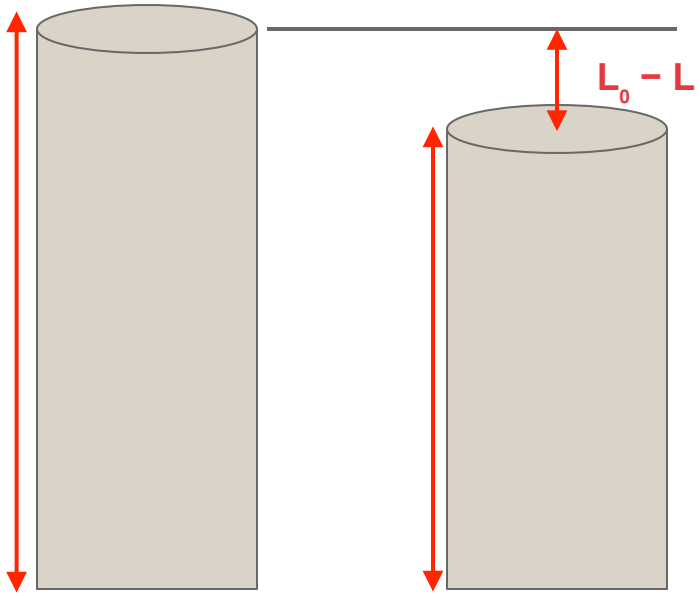
**Both shorten by 0.10 mm**




same shortening
different $L_0$

**Which specimen experiences the larger fractional change in length?**

# Strain normalizes the change in length




Axial Strain

$$\epsilon_a = \frac{L_0 - L}{L} = \frac{\Delta L}{L}$$

**Compression-positive convention**

**The reference length belongs in every strain definition.**

# What is the big picture here ?

Why do we make such observation?

- Equal shortening and different lengths → different strains.
- Equal stress and equal Young's modulus → equal strains, but different shortenings.

# What is the big picture here ?

Why do we make such observation?

- Equal shortening and different lengths → different strains.
- Equal stress → equal strains, but different shortenings.

Axial Strain

$$\epsilon_a = \frac{L_0 - L}{L_0} = \frac{\Delta L}{L_0}$$

# What is the big picture here ?

Why do we make such observation?

- Equal shortening and different lengths → different strains.
- Equal stress, because equal Young's modulus → equal strains, but different shortenings.

Axial Strain

$$\epsilon_a = \frac{L_0 - L}{L_0} = \frac{\Delta L}{L_0}$$

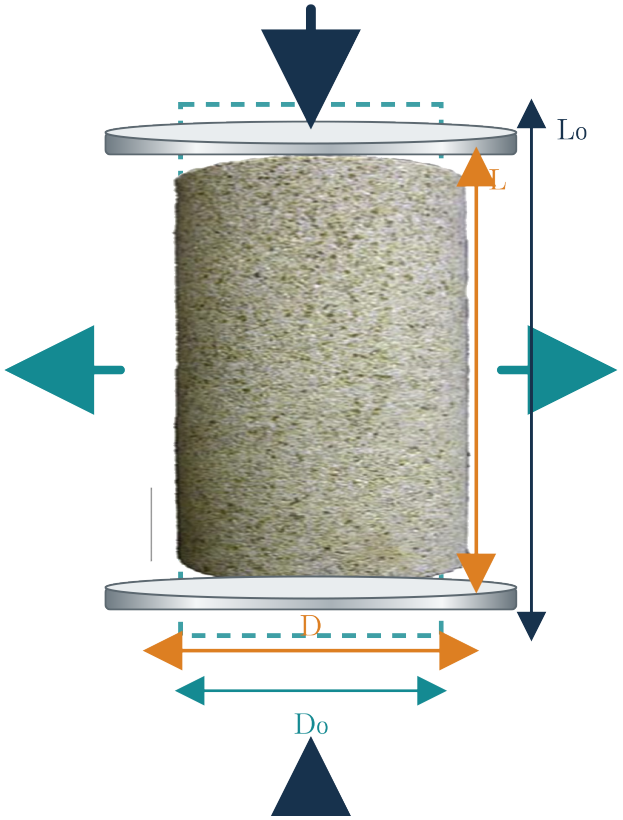
**Stress - strain relation**

**Next lecture**

$$\epsilon_a = \frac{\sigma_a}{E}$$

**E = Young's modulus**

# Deformation occurs in more than one direction




**Observed deformation**

- axial shortening
  $\Delta L = L_0 - L > 0$
- radial expansion
  $D > D_0$
  $\Delta D = D_0 - D < 0$

**$\varepsilon_a > 0$**

**$\varepsilon_r < 0$**

$$\epsilon_a = \frac{L_0 - L}{L_0} = \frac{\Delta L}{L_0}$$

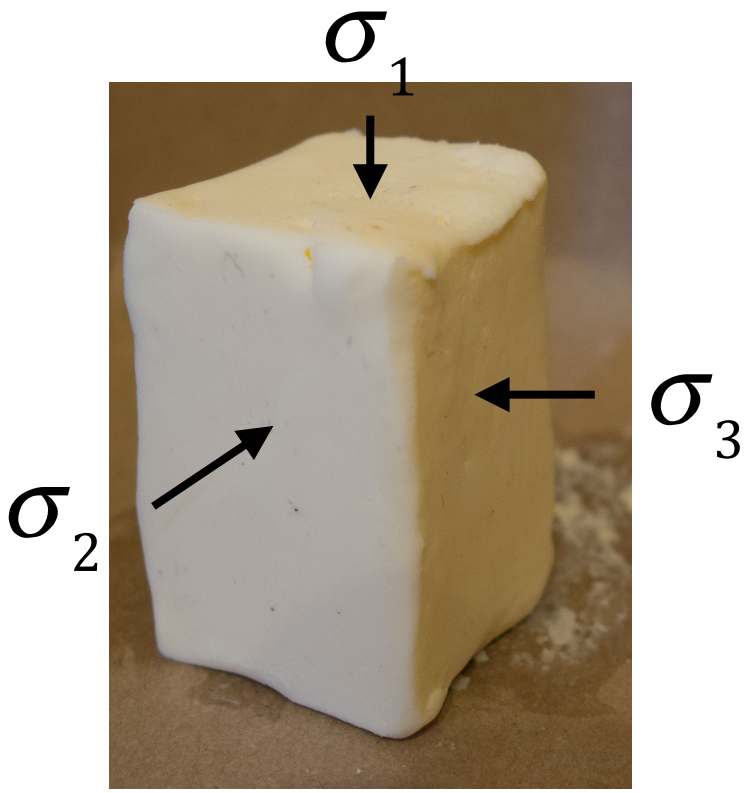
$$\epsilon_r = \frac{D_0 - D}{D_0} = \frac{\Delta D}{D_0}$$

For small strains and axisymmetric deformation:

$$\epsilon_v = \epsilon_a + 2\epsilon_r$$

**Can volume decrease while the specimen widens?**

# Principal stresses remove shear traction




**Recall stress assumption on Anderson theory**

$$\sigma_1 \geq \sigma_2 \geq \sigma_3$$

Three orthogonal principal directions

Nine components of the stress tensor

$$\begin{bmatrix} \sigma_x & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_y & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_z \end{bmatrix}$$

**Principal directions need not coincide with vertical and horizontal.**

# Uniaxial, hydrostatic and triaxial loading

Axial stress

Confining pressure




$\sigma_1 > 0$

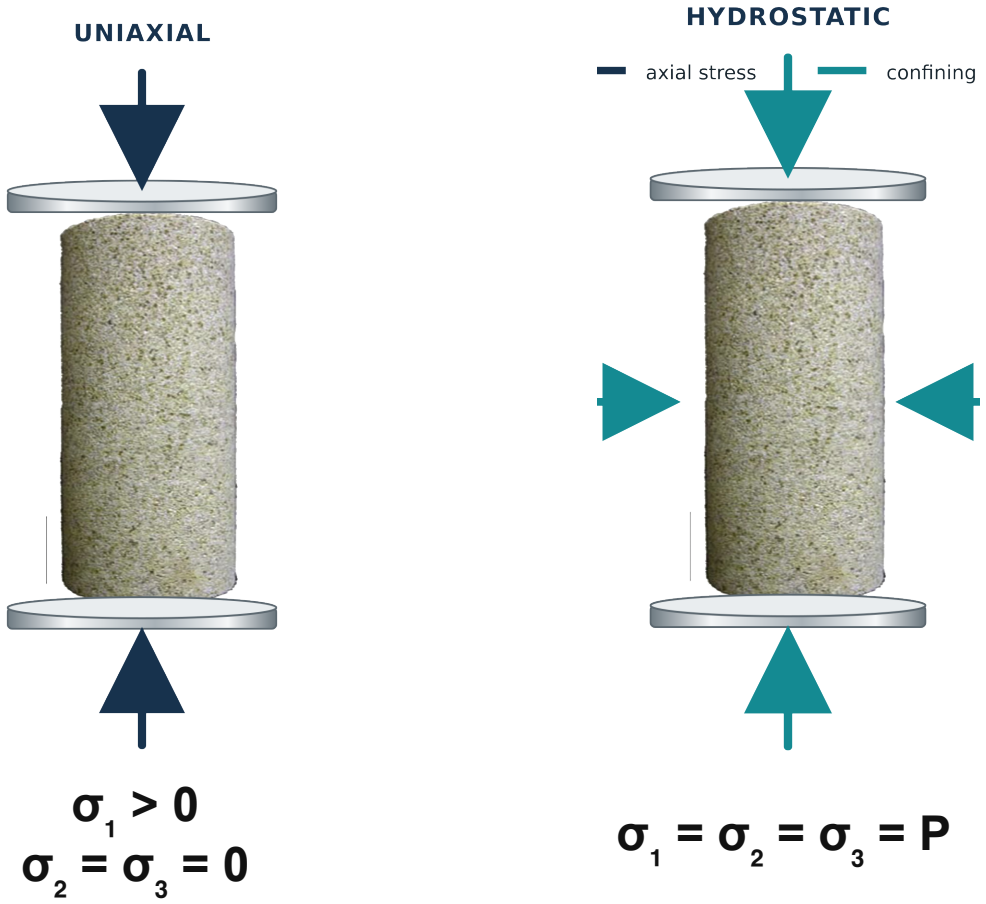
$\sigma_2 = \sigma_3 = 0$

**A free lateral surface sets lateral traction to zero, not lateral strain.**

# Uniaxial, hydrostatic and triaxial loading

Axial stress

Confining pressure




$\sigma_1 > 0$
$\sigma_2 = \sigma_3 = 0$

$\sigma_1 = \sigma_2 = \sigma_3 = P$

**A free lateral surface sets lateral traction to zero, not lateral strain.**

# Uniaxial, hydrostatic and triaxial loading

Activity 3

Axial stress

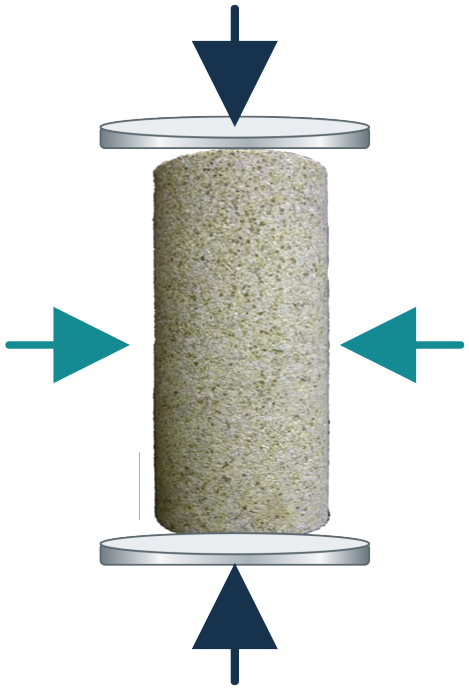
Confining pressure




$\sigma_1 > 0$
$\sigma_2 = \sigma_3 = 0$




$\sigma_1 = \sigma_2 = \sigma_3 = P$

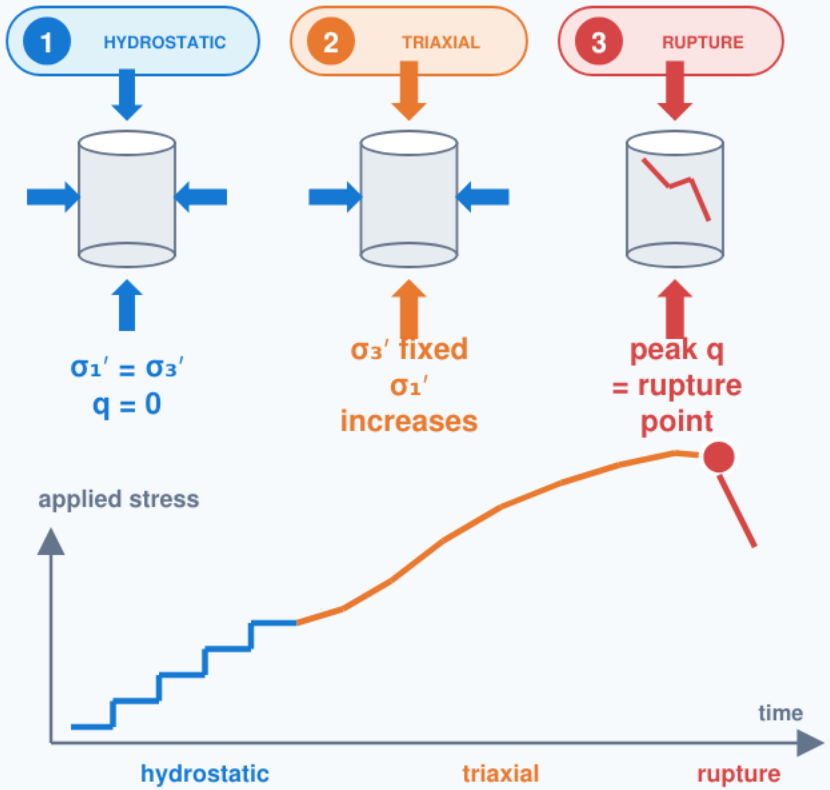



$\sigma_1 > \sigma_2 = \sigma_3 = Pc$

**A free lateral surface sets lateral traction to zero, not lateral strain.**

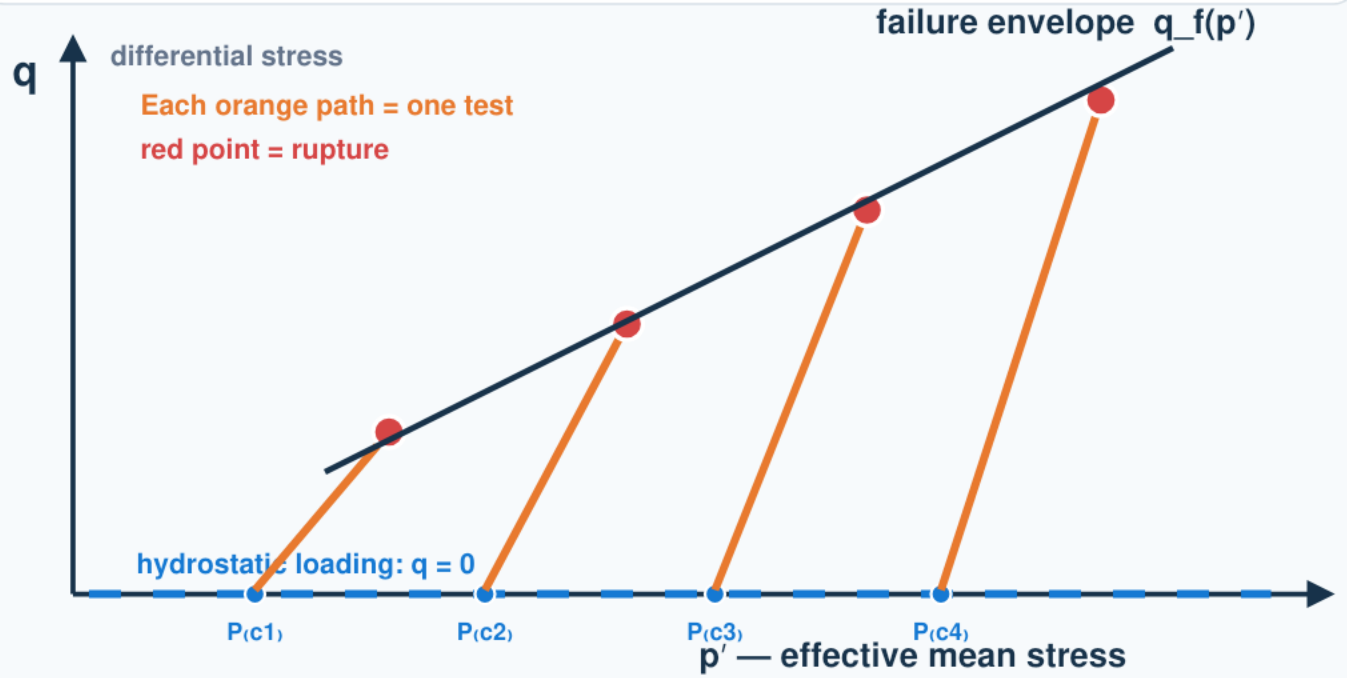
# How repeated triaxial tests build the p–q failure envelope

## ONE EXPERIMENT — follow the loading sequence




## SEVERAL EXPERIMENTS — retain the rupture point

$p' = (\sigma_1' + 2\sigma_3')/3$ $q = \sigma_1' - \sigma_3'$ at fixed $\sigma_3'$: $p' = \sigma_3' + q/3$

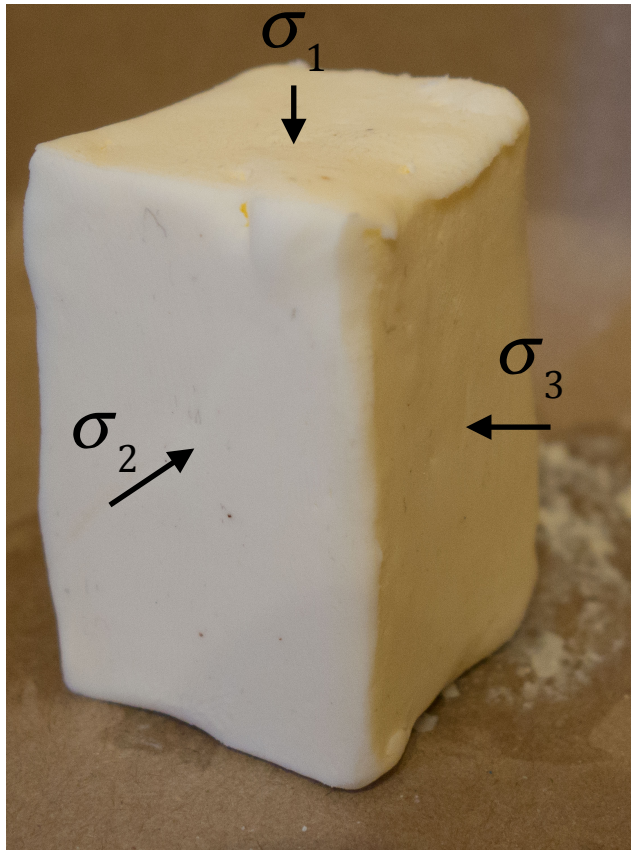


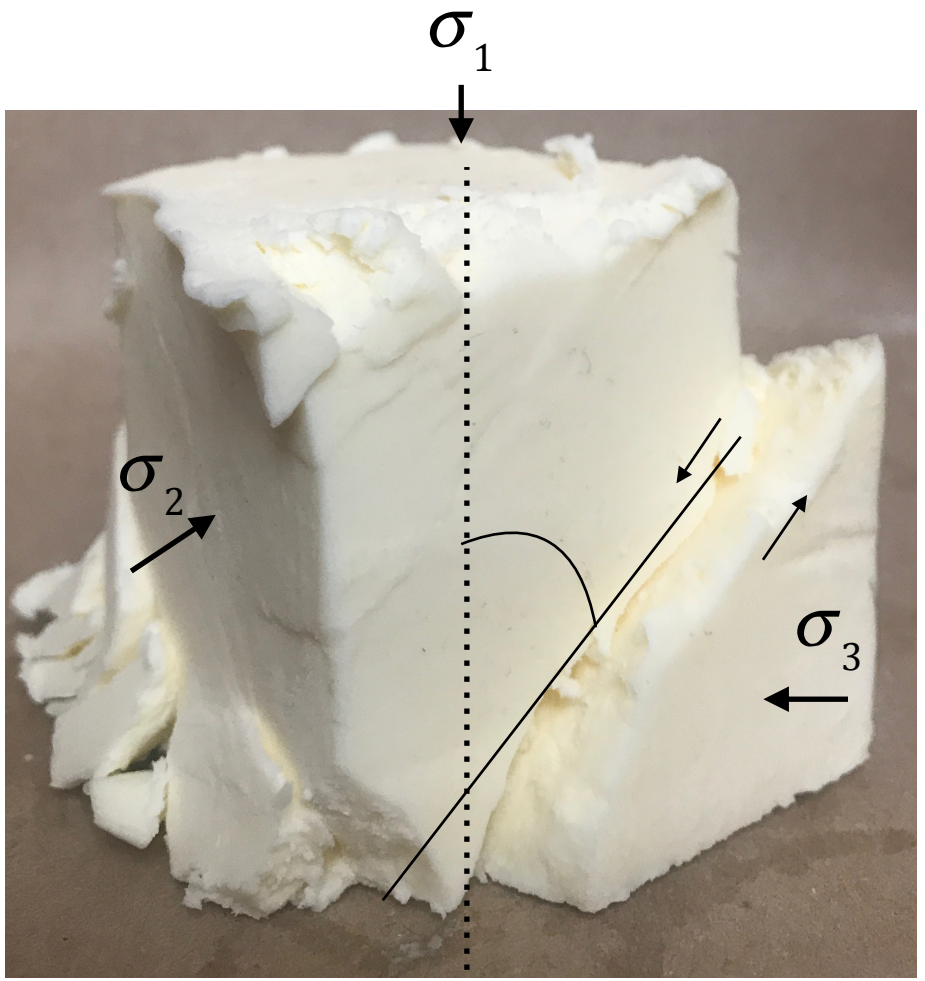

**ONE TEST** gives one rupture point. **REPEAT** at several confining pressures → **THE ENVELOPE EMERGES**

Adapted from Scuderi et al. (2015), G³, doi:10.1002/2015GC005932

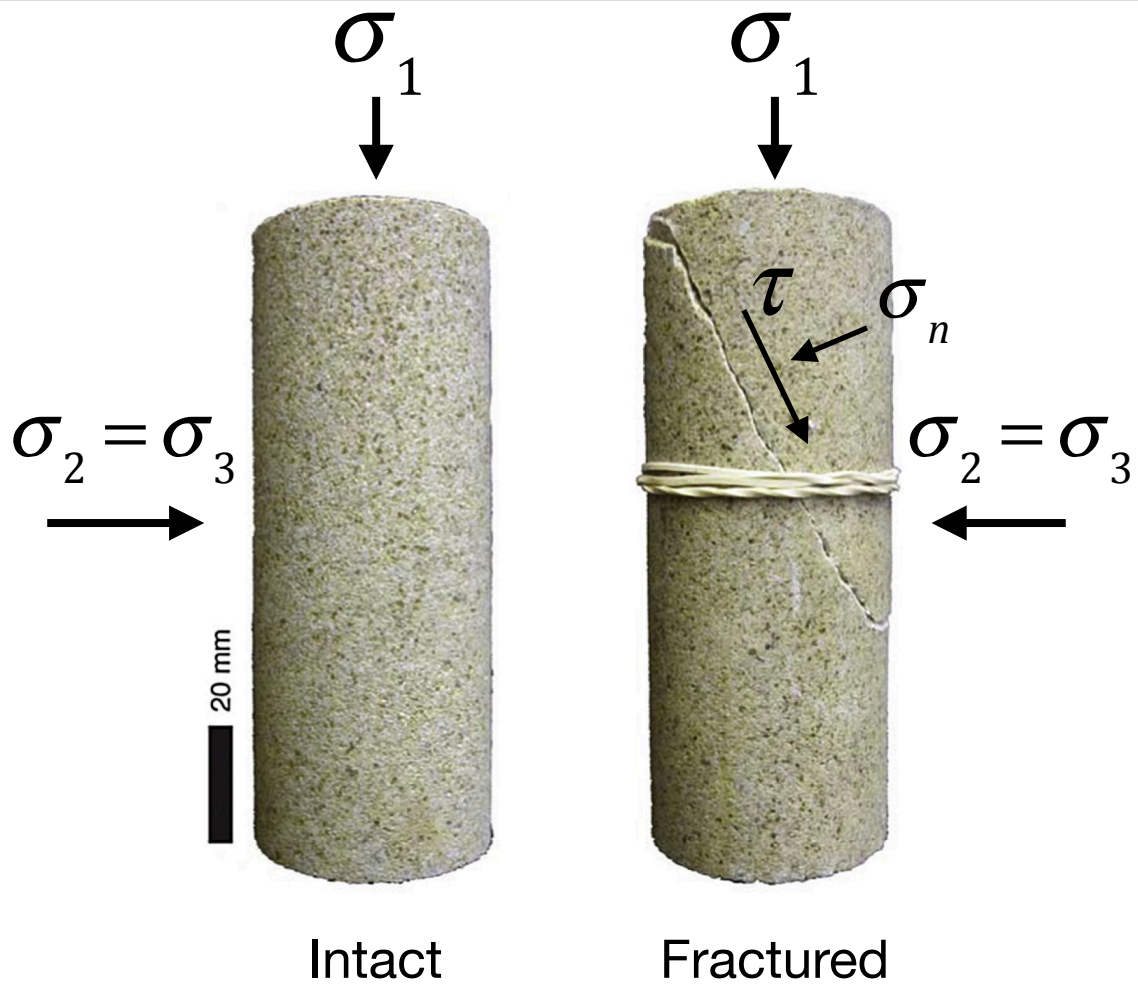
**How can we derive the stress conditions for the formation of a fracture/fault?**

**How can we characterise the stress field acting on the fault plane?**



# Stress acts across an imagined plane




**Normal stress $\sigma_n$**

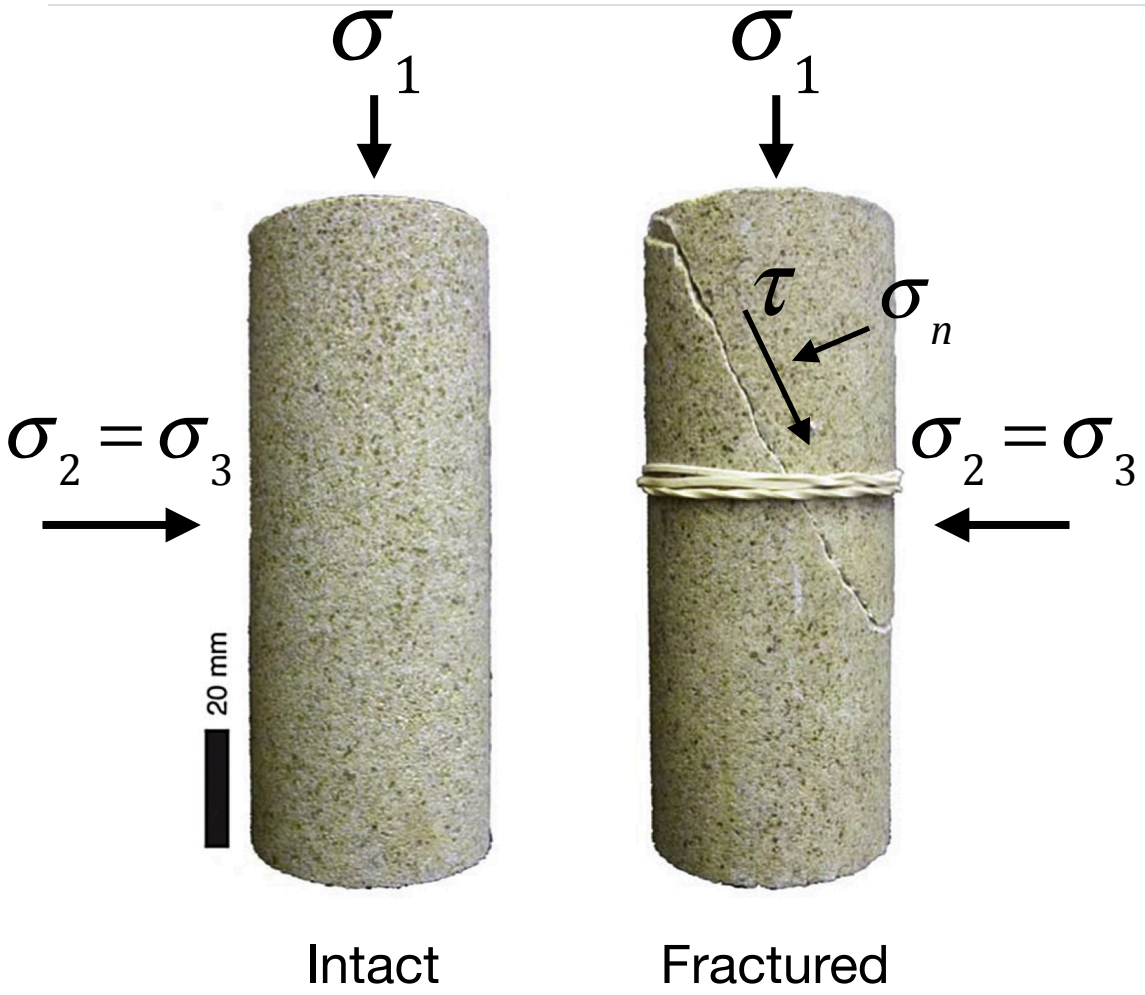
Perpendicular to the plane

**Shear stress τ**

Parallel to the plane

**The decomposition applies before a crack exists.**

# The inclined plane becomes a stress calculation




KNOWN

## Principal stresses

$\sigma_1 = 100MPa$

$\sigma_3 = 20MPa$

## Potential plane orientation

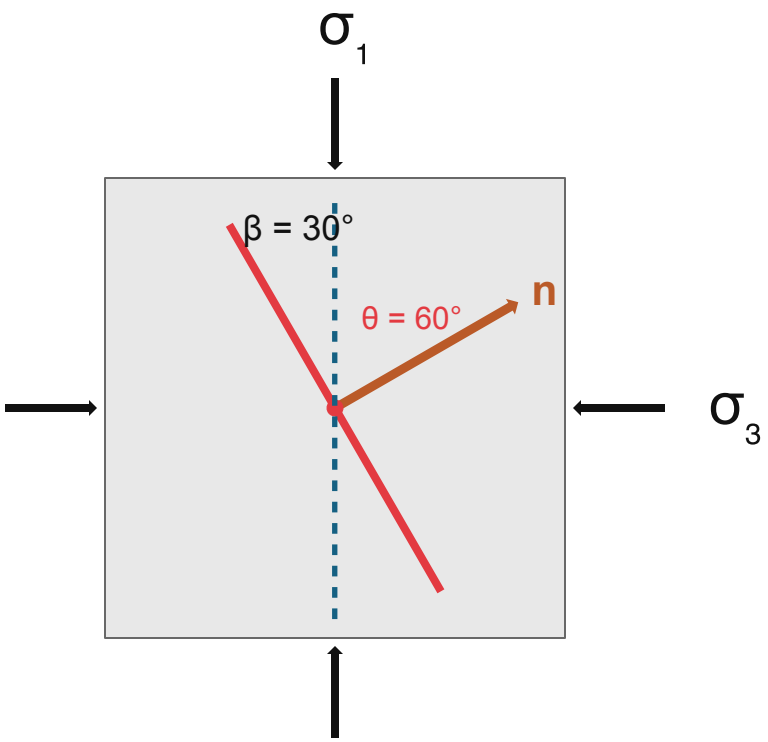
$\beta = 30^0$

CALCULATE

## Normal stress $\sigma_n$
## Shear stress τ

**Mark the plane normal before choosing an angle.**

# The angle between the plane normal and $\sigma_1$




**θ** Angle between n and $\sigma_1$

**β** Angle between the plane and $\sigma_1$

**$\theta + \beta = 90°$**

Here: $\theta = 60°$, $\beta = 30°$

**Every equation that follows uses this convention.**